



ENCOURAGING STEAM CAREERS AMONG HIGH SCHOOL STUDENTS

in the Indigenous Community of San Francisco Pichátaro, Michoacán

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INTRODUCTION

STEAM careers (Science, Technology, Engineering, Arts, and Mathematics) are essential for building a sustainable economy. Although the participation of women in these fields has increased, it remains insufficient compared to that of men. This gap is even more pronounced in indigenous communities, where structural, cultural, and economic barriers limit young women's access to information and educational opportunities.

Promoting the participation of indigenous women in STEAM careers is crucial for their economic autonomy, professional development, and personal and community empowerment. This must be done from a pluricultural and gender perspective that recognizes and values their identities and traditional knowledge.

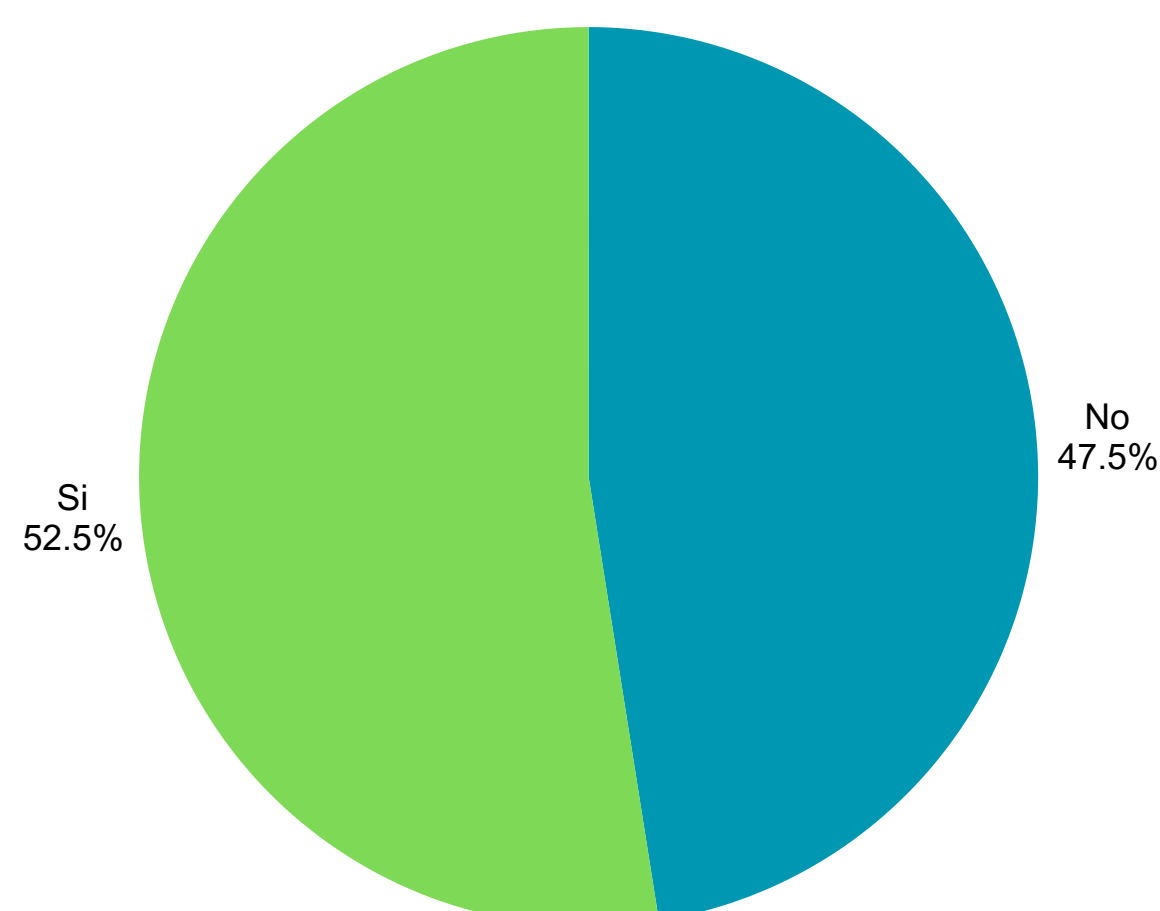
This project seeks to foster the interest of high school female students from San Francisco Pichátaro, Michoacán, in STEAM careers through an integrated program that includes diagnosis, raising awareness, and vocational guidance, and hands-on workshops. These activities aim to help participants recognize their potential and make informed decisions about their academic futures. The project will be carried out in three phases, beginning with data collection and raising awareness.



First-semester students completing the research questionnaire in the classroom at COBAEM Pichátaro Campus.

PRELIMINARY RESULTS

52.5% of the female students sample reported being aware of what STEAM areas are. Based on this result, an awareness stage was initiated to provide a solid foundation of knowledge, as well as to spark curiosity and interest for the upcoming phases of the program.



Question 13 of the research questionnaire: "Have you ever heard of or received any information about STEAM careers (Science, Technology, Engineering, Arts, and Mathematics)?"

ACKNOWLEDGMENTS:

Thank you to the high school, its teachers, and administrative staff, as well as to the students from all grade levels, for your time and support, which made it possible to successfully complete this first stage of the project.

REFERENCES

- Ebru Özdemir. (2023, January 5). Así es como la igualdad de género en áreas STEM puede favorecer una economía sostenible. Foro Económico Mundial. <https://es.weforum.org/stories/2023/01/asi-es-como-la-igualdad-de-genero-en-areas-stem-puede-favorecer-una-economia-sostenible/>
- Necesitamos más mujeres en carreras STEM. (2022, February 11). ONU Mujeres - América Latina Y El Caribe. <https://lac.unwomen.org/es/stories/noticia/2022/02/necesitamos-mas-mujeres-en-carreras-stem>

GENERAL OBJECTIVE

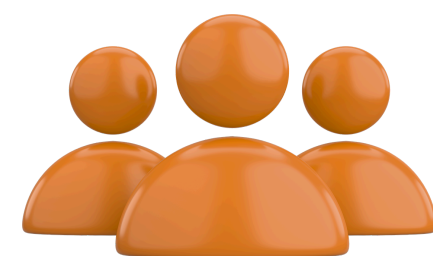
To promote the interest and participation of high school female students from the indigenous community of San Francisco Pichátaro, Michoacán, in STEAM fields (Science, Technology, Engineering, Arts, and Mathematics) through a comprehensive program with a gender and intercultural approach that integrates diagnosis, vocational guidance, and practical workshops.

SPECIFIC OBJECTIVE

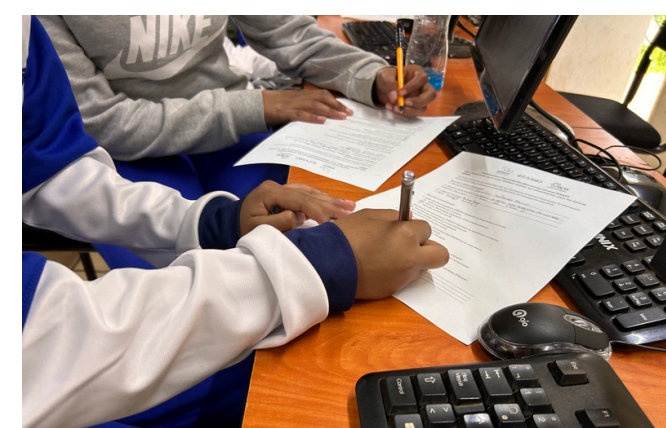
To assess the level of knowledge and interest of male and female students regarding STEAM careers and to identify the main barriers they face.

METHODS

This study employed an exploratory-descriptive approach aimed at identifying the level of knowledge possessed by students of the San Francisco Pichátaro High School regarding STEAM careers (Science, Technology, Engineering, Arts, and Mathematics).

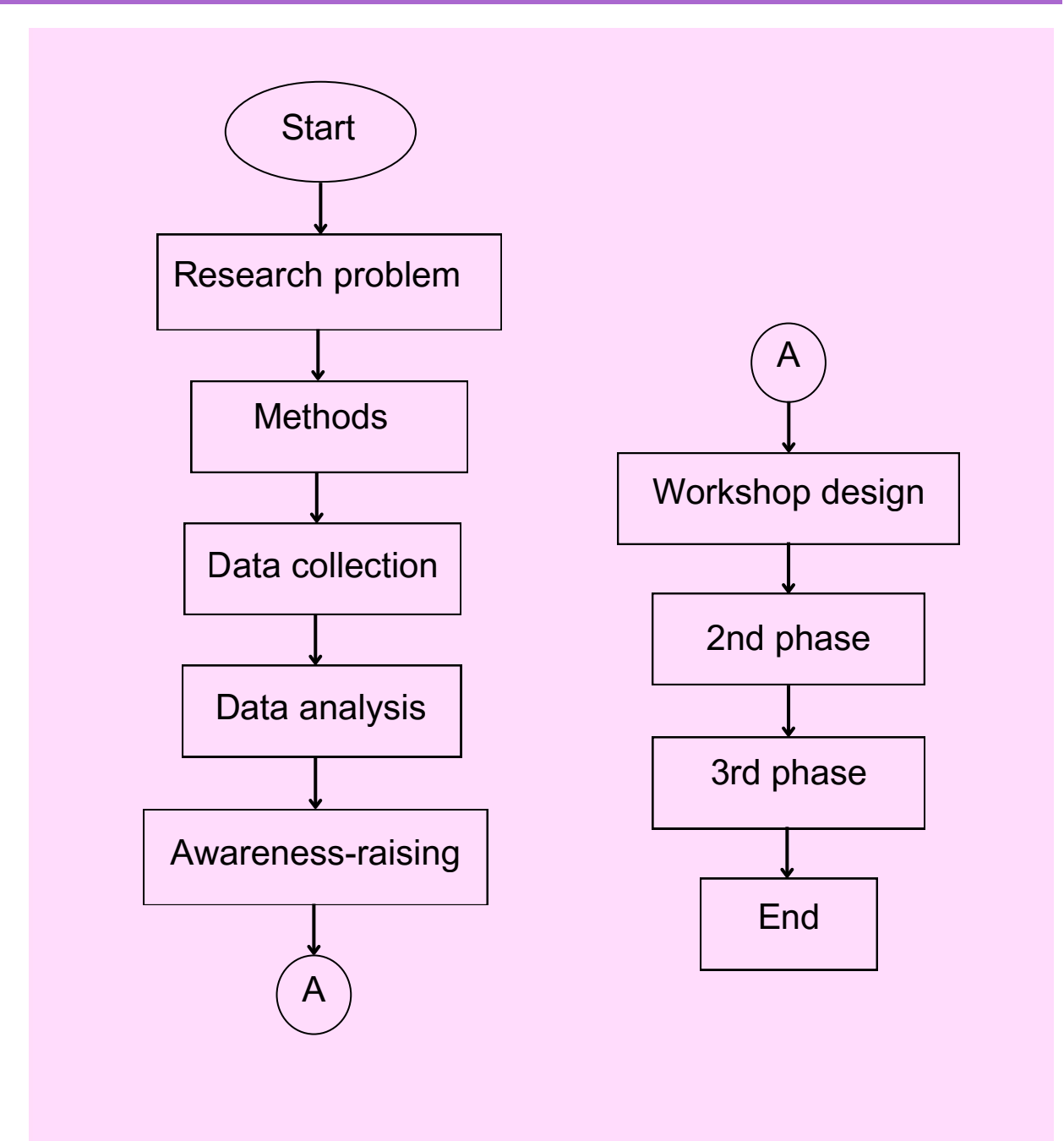


**80 women
52 men**



Fifth-semester students completing the research questionnaire.

21 preguntas



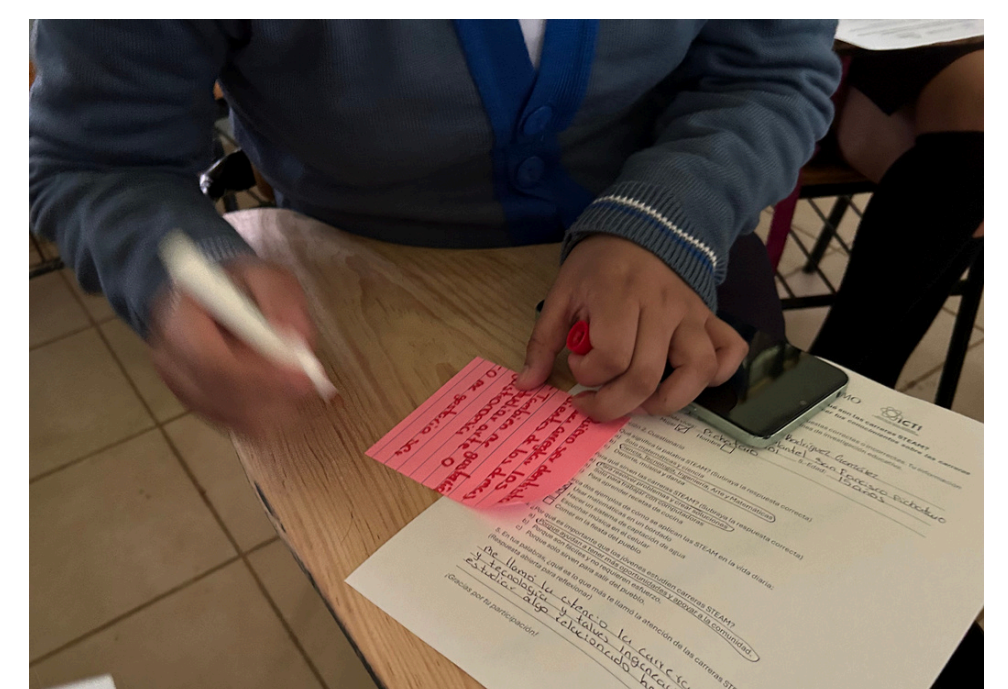
Flowchart illustrating the project process.

NEXT STEPS

- Meeting with parents: encourage the active participation of parents and guardians in supporting their daughters' educational processes.
- Collaborative work with the school: strengthen extracurricular activities and align efforts with the school's own projects.
- Content planning: use strategic educational content to maintain students' attention and participation throughout the program's activities.
- Focus group: create support networks, safe spaces, and workshops designed exclusively for the school's female students.
- Strategic alliances: establish partnerships with civil organizations, social enterprises, and local authorities to promote collaborative work..



Awareness-raising session about the project with teachers and school administrators.



Awareness activity through the presentation "What is STEAM?"

MORE INFO:

